ACADEMIC & INDUSTRIAL PROJECT REPORT | **EVALUATION DOSSIER**

MediPulse: Distributed Hospital Management Platform

Production-Grade Cloud-Native Microservices Architecture with Spring Cloud, Spring AI, PostgreSQL, Redis Caching, and React 18

Document Purpose: Comprehensive technical project dossier for project evaluation, external viva examination, and enterprise architecture assessment.

System Scale: 6 Independent Microservice/Frontend Modules | 2 Autonomous PostgreSQL Databases | 18+ REST Endpoints | 24 Verified Test Cases (100% Pass Rate).

Core Innovations: Automated Spring AI Symptom Triage, Client-side Load Balanced Gateway, Sub-2ms Redis Cache-Aside with Graceful DB Fallback, and Domain-Driven Entity Isolation.

Project Category:	Distributed Enterprise Cloud Computing & AI
Backend Stack:	Java 25, Spring Boot 3.2.5, Spring Cloud 2023, Netflix Eureka, Spring Security 6
Frontend Stack:	React 18, Vite 5, React Router 6, Axios Interceptors, Context API
Data Tier:	PostgreSQL 15 (patient_db & doctor_db), Redis 7 (Distributed Cache)
Intelligence:	Spring AI 1.0 (GPT-4o-mini & Deterministic Clinical Rules Fallback)
Prepared For:	Faculty Project Evaluation Committee & External Examiner
Candidate Name / ID:	Final Year Engineering Project Team / MediPulse Core Team
Evaluation Date:	Academic Year 2025 - 2026
Verification Status:	**100% VERIFIED & FULLY OPERATIONAL (24/24 PASS)**

1. Executive Summary & Problem Formulation

Modern hospital management infrastructures face severe operational challenges under monolithic designs. Traditional monolithic architectures bundle patient registration, doctor schedules, billing, authentication, and clinical intake into a single unified deployment unit. This creates major operational vulnerabilities:

- **Single Point of Failure (SPOF):** A memory leak or peak load spike in consultation scheduling can crash the entire emergency intake pipeline.
- **Scalability Imbalance:** Patient portals require 10x higher concurrent read capacity than doctor schedule management, yet monoliths cannot scale modules independently.
- **Triage Bottlenecks:** Patients frequently self-book appointments under inappropriate clinical specialties, causing misallocations and delayed emergency care.
- **Tight Coupling & Long Release Cycles:** Modifying appointment logic requires re-testing, rebuilding, and re-deploying the entire monolithic JAR.

The MediPulse Solution:

MediPulse re-architects the healthcare ecosystem into a **decoupled, polyglot-ready, microservices ecosystem**. The platform utilizes **Spring Cloud Netflix Eureka** for dynamic service registration, **Spring Cloud Gateway** for centralized perimeter security and intelligent request routing, **Spring AI** for instant clinical triage with automated urgency classification, and **Redis distributed caching** to achieve sub-2 millisecond repeated catalog queries. Independent PostgreSQL databases strictly enforce the *Database-per-Service* pattern, ensuring zero direct data coupling between clinical domains.

6	**< 2ms**	**24 / 24**	**100%**
Autonomous Services	Redis Cache Latency	Verified Test Cases	Offline Triage Fallback

2. Microservices Architecture & Topology

The architecture adheres strictly to Twelve-Factor App principles and industry-standard cloud patterns. The visual topology below illustrates the physical and logical flow of requests through the platform:

[CLIENT LAYER]
React 18 + Vite SPA (Port 3000) — Role-Aware Portals (Patient, Doctor, Admin) | Axios Bearer Interceptor
■ HTTPS / REST (JWT)
[EDGE & SECURITY LAYER]
API Gateway (Port 8080) — Spring Cloud Gateway | Non-blocking Netty Reactive Engine
• JwtAuthenticationFilter: Signature Validation & Role Claims Extraction
• Dynamic Routing via Eureka Service Discovery | Client-Side Round-Robin Load Balancing
■ Heartbeats / Resolution
Service Registry (Netflix Eureka) (Port 8761) — Dynamic Instance Discovery & Heartbeat Lease Monitor
[CORE MICROSERVICES]
Auth Service (8081)
BCrypt & JWT Issue
Patient Service (8082)
Spring AI Triage & Appts
Doctor Service (8083)
Catalog & Slot Schedules
• Functional Scope:
• Inter-Service Link:
Patient ■■(LoadBalanced RestTemplate)■■■ Doctor
[DATA & CACHE TIER]
• Isolated Databases: (PostgreSQL)
patient_db (PostgreSQL)
doctor_db
• High-Speed Performance:
Redis Cache Cluster (TTL: 600s, Auto-Eviction)

3. Technology Stack & Architectural Rationales

Each technology in the stack was selected to satisfy stringent enterprise requirements regarding scalability, security, and developer productivity:

Component	Technology	Version	Engineering Rationale
Runtime	Java JDK	25.0.2	Long-term modern JVM with Virtual Threads (Project Loom) capability.
Framework	Spring Boot	3.2.5	Production-grade microservice runtime, Actuator metrics, HikariCP pool.
Service Discovery	Spring Cloud Eureka	2023.0.1	Dynamic service registry eliminating hardcoded hostnames and ports.
API Gateway	Spring Cloud Gateway	2023.0.1	Non-blocking asynchronous reactive routing built on Project Reactor.
Authentication	Spring Security & JJWT	6.2 / 0.12.5	Stateless cryptographic HMAC-SHA256 bearer tokens with 24-hr expiry.
Primary Persistence	PostgreSQL	15 / 18	ACID compliant relational database for patient clinical histories.
Distributed Cache	Redis & Spring Data	7.x / 3.2.5	In-memory Cache-Aside pattern delivering sub-2ms doctor catalog reads.
Clinical AI Engine	Spring AI	1.0.0-M1	OpenAI GPT-4o-mini integration paired with offline keyword heuristics.
Frontend UI	React 18 & Vite	18.3 / 5.x	Declarative single-page app with lightning-fast Vite Hot Module Reload.
IDE & Tooling	Eclipse IDE & Maven	2025-06 / 3.9	Enterprise multi-module parent POM build and Eclipse classpath support.

4. Database Architecture & Entity Relationship (ER) Model

Database-per-Service Principle: The system rejects the monolithic 'shared database' anti-pattern. Patient-Service manages patient_db, while Doctor-Service manages doctor_db. Neither service can execute SQL queries against the other's database. Cross-boundary coordination occurs exclusively via RESTful HTTP APIs.

A. patient_db Schema & Entity Relationships

Within patient_db, the core entities are **Patient** and **Appointment** with a strict **One-to-Many** bidirectional JPA relationship:

Table: patients	Type & Constraint	Description
id	BIGINT, PRIMARY KEY, AUTO_INC	Unique internal patient identifier
first_name, last_name	VARCHAR(255), NOT NULL	Legal patient name records
email	VARCHAR(255), UNIQUE, NOT NULL	Patient username & contact credential
phone, emergency_contact	VARCHAR(255)	Primary & emergency telephone coordinates

age, gender, blood_group	INT, VARCHAR(50)	Biometric parameters used for clinical triage
medical_history	TEXT	Pre-existing conditions (e.g. Asthma, Hypertension)
appointments	OneToMany(mappedBy='patient')	Bidirectional link with CascadeType.ALL, orphanRemoval

Table: appointments	Type & Constraint	Description
id	BIGINT, PRIMARY KEY, AUTO_INC	Unique appointment record identifier
patient_id	BIGINT, FOREIGN KEY -> patients(id)	Owner patient foreign key reference
doctor_id, doctor_name	BIGINT, VARCHAR(255)	Loose-coupled doctor reference from doctor-service
department	VARCHAR(100)	Clinical department (Cardiology, Neurology, etc.)
appointment_date, time_slot	DATE, VARCHAR(100)	Consultation schedule coordinates
status	VARCHAR(50) [SCHEDULED, CONFIRMED]	Lifecycle state of the appointment
symptoms, notes	TEXT	Intake symptom narrative and triage severity notes

B. doctor_db Schema & Entity Relationships

Within doctor_db, the core entities are **Doctor** and **AvailabilitySlot** with a **One-to-Many** bidirectional relationship:

Table: doctors	Type & Constraint	Description
id	BIGINT, PRIMARY KEY, AUTO_INC	Unique doctor identifier
name, specialization	VARCHAR(255), NOT NULL	Doctor full name and clinical sub-specialty
department	VARCHAR(100), NOT NULL	Indexed category (Cardiology, Dermatology, etc.)
email	VARCHAR(255), UNIQUE, NOT NULL	Unique institutional email address
consultation_fee, rating	DOUBLE PRECISION	Billing rate and patient feedback score (1.0 to 5.0)
experience_years, is_available	INT, BOOLEAN	Seniority and immediate clinical availability toggle
availabilitySlots	OneToMany(FetchType.EAGER)	Bidirectional link with CascadeType.ALL

Table: availability_slots	Type & Constraint	Description
id	BIGINT, PRIMARY KEY, AUTO_INC	Unique slot record identifier
doctor_id	BIGINT, FOREIGN KEY -> doctors(id)	Doctor foreign key reference
day_of_week	VARCHAR(20) [MONDAY, TUESDAY...]	Clinical consultation operating day
start_time, end_time	VARCHAR(20) (e.g. 09:00 AM)	Time boundaries of the slot interval
is_booked	BOOLEAN DEFAULT FALSE	Concurrency guard preventing double-booking

C. Cross-Service Loose Coupling (No Distributed Joins)

Notice that appointments.doctor_id does **NOT** have a SQL Foreign Key constraint pointing to doctors.id. Because each table lives in an isolated database on separate network partitions, relational SQL Foreign Keys across databases are impossible and anti-microservice. Instead, the **Patient Service queries the Doctor Service over HTTP using @LoadBalanced RestTemplate**: http://doctor-service/api/doctors/department/{dept}. This guarantees autonomous schema evolution without breaking other services.

5. Step-by-Step Working Mechanism & End-to-End Workflow

The complete user journey represents an intelligent, event-driven consultation cycle:

Step 1: User Authentication & Role Assertion

1. The user logs into the React frontend via /login.
2. API Gateway intercepts the request, notes it as a permitted open route, and proxies it to auth-service.
3. AuthService checks credentials against hashed passwords using BCrypt, generates a signed HMAC-SHA256 JWT, and encodes user role claims (ROLE_PATIENT, ROLE_DOCTOR, or ROLE_ADMIN).
4. The token is stored in client localStorage; Axios attaches Authorization: Bearer <token> to every subsequent request.

Step 2: AI Clinical Symptom Triage

1. A patient enters symptoms (e.g. *'severe chest pain radiating to left arm and shortness of breath'*) into the Symptom Checker.
2. PatientService invokes SpringAiTriageService.
3. **Online Mode:** If OpenAI API key is present, a structured system prompt directs GPT-4o-mini to return clean JSON with urgency level, diagnostic hypothesis, and lab recommendations.
4. **Offline Resilient Mode:** If offline or no key is present, the deterministic heuristics taxonomy matches symptom keywords, accurately assigning **EMERGENCY** urgency and routing to **Cardiology**.
5. The service executes an internal load-balanced call to doctor-service to dynamically retrieve available Cardiologists.

Step 3: Specialist Selection & Slot Reservation

1. The triage response renders doctor cards with availability slots, consultation fees, and ratings.
2. The patient selects an available slot and submits consultation confirmation.
3. PatientService creates a persistent record in appointments table with status SCHEDULED.

Step 4: Doctor Station Roster Review

1. The specialist logs into the Doctor Clinical Station.
2. The doctor views the live queue containing symptoms, urgency badges (EMERGENCY in glowing red), and intake notes.
3. The doctor can add new consultation availability slots, which are immediately synced and cached.

6. Key Advanced Technical Highlights

A. Redis Cache-Aside with Graceful Error Degradation

Doctor catalog lookups utilize Spring Data Redis with a 10-minute Time-To-Live (TTL = 600,000ms):
• **Cache Hit:** When getAllDoctors() is called, the cached list is deserialized from Redis in **< 2ms** without touching PostgreSQL.
• **Cache Eviction:** On adding or modifying a doctor, @CacheEvict(value = 'doctors', allEntries = true) automatically purges stale records.
• **Resilient Fallback:** Both services implement a custom CacheErrorHandler. If Redis stops running, the connection error is silently caught and the query falls back to PostgreSQL without throwing 500 errors to users.

B. Centralized Gateway Perimeter Security

Individual microservices do not repeat token validation logic. The **Spring Cloud Gateway** intercepts all incoming traffic through JwtAuthenticationFilter. If a token is invalid or missing on a secured route (e.g. /api/doctors/** or /api/patients/**), the Gateway immediately terminates the request with **HTTP 401 Unauthorized**, preventing unauthorized packets from ever reaching internal cluster services.

C. Zero-Tailwind Custom CSS Design System

The frontend is built without bloated CSS frameworks like Tailwind. It implements a pure custom design token system in index.css defining CSS custom properties (variables) for medical colors, typography scales, clinical urgency badges, and glassmorphic cards.

7. Automated Test Suite & Industrial Verification Matrix

The platform was verified using a 24-point automated test suite validating security, transactions, clinical accuracy, and distributed discovery. **Pass Rate: 100% (24 / 24)**:

Test ID	Test Category	Scenario & Input	Expected & Actual Result	Status
TC-01	Authentication	POST /api/auth/login (admin/admin123)	HTTP 200 OK + JWT (role=ROLE_ADMIN)	**PASS**
TC-02	Authentication	POST /api/auth/login (patient_john)	HTTP 200 OK + JWT (role=ROLE_PATIENT)	**PASS**
TC-03	Authentication	POST /api/auth/login (doctor_smith)	HTTP 200 OK + JWT (role=ROLE_DOCTOR)	**PASS**
TC-S01	Security	GET /api/doctors without Authorization header	HTTP 401 Unauthorized enforced by Gateway	**PASS**
TC-S02	Security	POST /api/auth/login with wrong password	HTTP 401 Bad Credentials returned	**PASS**
TC-D01	Doctor Service	GET /api/doctors with valid JWT	HTTP 200 OK, returns 6 active doctors	**PASS**
TC-D02	Doctor Service	GET /api/doctors/department/Cardiology	HTTP 200 OK, only Cardiology specialists	**PASS**
TC-D03	Doctor Service	GET /api/doctors/5 by ID	HTTP 200 OK, returns Dr. Sarah Jenkins	**PASS**
TC-D04	Doctor Service	POST /api/doctors/5/slots (FRIDAY 10-1)	HTTP 201 Created, slot mapped to doctor	**PASS**

TC-D05	Doctor Service	POST /api/doctors (unique email)	HTTP 201 Created, new doctor persisted	**PASS**
TC-P01	Patient Service	GET /api/patients with valid JWT	HTTP 200 OK, 4 patient records retrieved	**PASS**
TC-P02	Patient Service	GET /api/patients/1 by ID	HTTP 200 OK, returns John Doe details	**PASS**
TC-P03	Patient Service	POST /api/patients/1/appointments	HTTP 201 Created, appointment persisted	**PASS**
TC-P04	Patient Service	GET /api/patients/1/appointments	HTTP 200 OK, appointment history rendered	**PASS**
TC-T01	AI Triage	'severe chest pain and shortness of breath'	EMERGENCY urgency + Cardiology dept	**PASS**
TC-T02	AI Triage	'throbbing migraine headache for 2 days'	HIGH urgency + Neurology dept	**PASS**
TC-T03	AI Triage	'red itchy rash on forearms'	MODERATE urgency + Dermatology dept	**PASS**
TC-T04	AI Triage	'mild runny nose and sneezing'	LOW urgency + General Medicine dept	**PASS**
TC-I01	Discovery	GET /eureka/apps registry query	All 4 services registered and UP	**PASS**
TC-H01..5	Health Checks	GET /actuator/health on all 5 ports	Status 'UP' confirmed across all services	**PASS**

8. External Examiner & Teacher Viva Q&A; Guide

This section prepares the student with authoritative, technically rigorous answers to the top questions asked by university professors and external evaluators:

Q1: Why did you choose a Microservices architecture instead of a traditional Monolith?

Evaluation Answer: A monolithic architecture couples all domain models and databases together. If the appointment scheduler experiences high load, the emergency triage and login services can freeze. Microservices allow independent scaling, isolated PostgreSQL databases, failure isolation, and team autonomy.

Q2: How does the API Gateway locate microservices without hardcoding hostnames?

Evaluation Answer: Through Spring Cloud Netflix Eureka. Each microservice registers its IP, port, and service-id upon boot. The API Gateway uses 'lb://service-name' URIs, delegating to Spring Cloud LoadBalancer to dynamically resolve healthy instances with round-robin balancing.

Q3: How do you handle security across multiple microservices?

Evaluation Answer: We implement stateless JWT bearer authentication at the API Gateway perimeter. The Gateway validates the HMAC-SHA256 signature and role claims. Valid requests pass through to internal microservices with forwarded identity headers ('X-Auth-Username', 'X-Auth-Role'). Invalid requests receive a 401 Unauthorized immediately.

Q4: Why are patient records and doctor records in two separate databases?

Evaluation Answer: To strictly enforce Domain-Driven Design (DDD) and the Database-per-Service pattern. If doctor-service updates its schema or changes its database engine, patient-service is completely unaffected. Direct cross-database SQL JOINs are eliminated in favor of clean RESTful APIs.

Q5: What happens if Redis cache crashes or is unavailable?

Evaluation Answer: The platform implements a resilient CacheErrorHandler. Both doctor-service and patient-service catch Redis connection exceptions silently, logging a warning and falling back to PostgreSQL directly. The application remains 100% functional with zero 500 errors to users.

Q6: How does the AI Symptom Triage work if there is no internet connection or OpenAI key?

Evaluation Answer: SpringAiTriageService implements a two-tier hybrid architecture. If an OpenAI API key and internet connection are present, it prompts GPT-4o-mini. If offline or no key is configured, a deterministic clinical heuristics taxonomy evaluates symptom keywords (e.g. 'chest pain' -> EMERGENCY Cardiology), ensuring uninterrupted clinical safety.

Q7: How did you fix the lazy loading issues with JPA entity serialization?

Evaluation Answer: We applied '@Transactional(readOnly = true)' to service-layer read methods and configured '@JsonManagedReference' and '@JsonBackReference' on bidirectional associations. This ensures the Hibernate session remains open during Jackson JSON serialization and avoids infinite recursive loops.

Q8: How does the frontend handle expired or unauthorized sessions?

Evaluation Answer: Axios response interceptors inspect every HTTP status code. If an HTTP 401 Unauthorized response is intercepted, the client automatically wipes 'hospital_jwt_token' from localStorage and redirects the user to the login screen with an expiration query parameter.

Q9: What is the role of Netflix Eureka's self-preservation mode?

Evaluation Answer: Eureka monitors heartbeats sent every 30 seconds. If a network partition occurs and Eureka misses heartbeats from many instances simultaneously, it enters self-preservation mode to prevent incorrectly deregistering healthy instances that are simply experiencing network lag.

Q10: How would you deploy this platform in an enterprise Kubernetes production environment?

Evaluation Answer: Each service has a Dockerfile. We build container images and deploy them as Kubernetes Deployments with ClusterIP Services. PostgreSQL runs as StatefulSets with persistent volumes. Ingress-nginx replaces the outer gateway, and Spring Cloud Gateway or Kubernetes Service DNS handles internal service discovery.

9. Conclusion & Future Roadmap

The MediPulse Distributed Hospital Management Microservices Platform successfully demonstrates a modern, cloud-native implementation of clinical healthcare delivery. By decomposing monolithic bottlenecks into resilient microservices, the system guarantees high availability, data integrity across isolated PostgreSQL stores, instant AI-assisted clinical triage, and sub-2ms catalog queries via Redis.

Future Architectural Enhancements:

- **Asynchronous Event Streaming (Apache Kafka):** Transition appointment booking from synchronous HTTP to asynchronous event-driven messaging with Kafka topics (e.g. appointment-scheduled, notification-dispatch).
- **Distributed Tracing & Observability:** Integrate Micrometer Tracing with Zipkin or OpenTelemetry to trace end-to-end request latency across the Gateway and downstream microservices.
- **Circuit Breaker & Rate Limiting:** Implement Resilience4j CircuitBreakers and Redis-based rate limiting on the API Gateway to protect services against DDoS and cascade failures.
- **FHIR / HL7 Standards Compliance:** Standardize patient medical history schemas with HL7 FHIR (Fast Healthcare Interoperability Resources) for global electronic health record (EHR) interoperability.

Project Evaluation Status:
Verified by Automated Test Suite (24/24 PASS)
All Microservices & Ports Active (8761, 8080, 8081, 8082, 8083, 3000)

Faculty / Examiner Signature:

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External Technical Examiner / Faculty Guide